Video Dehazing and Object Detection using Deep Learning

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***Abstract*—Computer vision systems have become very popular among modern IoT applications, autonomous driving, and security systems. The new generation of wireless networks offers high speeds and low-latency support for the real-time applications of these systems. However environmental challenges like haze and fog significantly degrade their performance. This paper presents a real-time system that combines advanced video dehazing and object detection to address this limitation. The proposed method processes video footage at dynamic rates, treating each frame as an individual hazy image. ConvIR, a neural network model built on encoder-decoder architecture and residual layers, is applied to enhance image clarity by removing atmospheric noise. The dehazed images are then passed through YOLOv5, a state-of-the-art object detection model, which accurately identifies and classifies objects. Our approach is backed by exceptional performance metrics, achieving an object detection accuracy of 0.996 and a peak signal-to-noise ratio (PSNR) of 34.5 in the dehazing phase. This integration ensures superior performance in adverse weather conditions, enabling enhanced visibility and robust object detection. The proposed system holds immense potential for real-world applications such as autonomous driving and surveillance, offering improved safety and reliability in low-visibility scenarios.**

***Index Terms*—Video dehazing, Object Detection, Deep Learning (DL), CONVIR, YOLOv5**

I. Introduction

A. Problem Statement

In today's world, technological advancements in the convergence of cloud computing and artificial intelligence have enabled groundbreaking innovations across industries. This synergy and real-time big data analysis have fueled the hype surrounding autonomous driving applications. These systems deliver a safe and optimal response by processing sensor data and traffic information. However, this requires strong preprocessing steps to factor in environmental noise. Innovations in video dehazing and object detection are thus given vital importance to ensure the robust functioning of the system. Environmental noise in the form of haze and fog in the atmosphere poses a significant challenge to computer vision systems such as autonomous driving, surveillance, and environmental monitoring. To address this, video dehazing is incorporated which reconstructs a clear image from a degraded video by taking frames per second. This task combines image processing and deep learning techniques to enhance visual quality and extract useful information from deteriorated visual data [1].

Traditional image dehazing techniques studied physical models of atmospheric scattering and identified features to estimate and remove haze. However, their results underperformed in dynamic scenarios. Recent developments leverage deep learning, particularly convolutional neural networks (CNNs) and generative models, directly learn complex spatial and temporal features from data. They are expected to address the problems caused by adverse weather conditions. [2].

For a real-time application, the complexity of the model increases while ensuring smooth transition across consecutive frames. It introduces a loss factor that needs to be incorporated while reconstructing the dehazed image. Following the removal of atmospheric noise, the system needs to understand its environment by identifying its neighboring objects. This involves the object detection step. Detecting objects remains one of the most fundamental and challenging aspects of computer vision and image understanding applications [3]. There have been significant advancements in object detection with the utilization of deep neural networks. It involves image classification through localization and segmentation. Each instance in an object is to be identified and assigned a label to a specific class from a wide range of predefined classes. DL

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object detection models using convolutional neural networks and transformers are now playing a key role in the evolution of this domain. They can provide vital information for the semantic understanding of images and videos.

B. Major Contributions

The major contributions of this paper are as follows:

- We propose a real-time two-stage video processing pipeline integrating ConvIR-based dehazing with YOLOv5 object detection.
- We demonstrate that dehazing video frames prior to object detection significantly improves detection performance in hazy conditions.
- We utilize real-world video data extracted from the MARS dataset, with custom preprocessing and frame extraction based on native frame rates.
- Extensive experiments show that our approach outperforms directly applying object detection on hazy frames, highlighting its suitability for safety-critical applications such as autonomous driving and surveillance.

C. Applications

1) AI-assisted traffic signals: Video dehazing and object detection are the key technologies involved in AI-powered traffic signals. They assess the traffic load on each side of the junction and make a real-time judgment of which side traffic should flow and for how long. This technology is a far more efficient way to tackle traffic congestion and also optimizes the fuel consumption of vehicles. The dehazing model allows the system to adapt to adverse weather conditions such as fog, rain, and haze. The object detection model analyses the video feed to identify vehicles, pedestrians, and other entities present on the road. This is further expanded to surveillance cameras which can capture the license plate of vehicles that violate traffic rules. By integrating these technologies, AI-assisted traffic systems efficiently manage traffic, and predict congestion patterns, which can be useful insights for urban planners. For example, in Dammam, Saudi Arabia [4], such systems are envisioned to alleviate congestion, reduce emissions, and improve road safety. The adaptive nature of AI allows these systems to evolve with increasing urbanization and the development of a utopian world.

2) Self-driving vehicles: One of the most popular innovations today is autonomous vehicles (AV) which is one of the main applications of video dehazing and object detection. The technology ensures safe and efficient navigation, especially in adverse weather conditions. Upon reconstructing the clear image ahead of the vehicle, the model identifies obstacles in its vicinity and maps its relative distance. It then makes an informed decision to continue on its path without colliding with incoming traffic. The model also reads traffic signals before making its decision.

Traditional object detection methods, such as Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradient (HOG), depend on pre-trained features to identify objects [5]. These methods are accurate in ideal conditions but struggle in dynamic environments. Fog, haze, snow, and other impurities distort the sensor data drawn from the cameras installed in the AV. These systems will not be able to make the right decision in such situations. Deep learning (DL) algorithms, on the other hand, extract features and adapt to varying conditions. Convolutional neural networks (CNNs) such as YOLO (You Only Look Once) and Faster R-CNN are being adopted in AVs. YOLO excels in real-time applications, while Faster R-CNN provides high precision, although it is computationally demanding. By integrating reliable object detection systems, AVs can navigate complex traffic scenarios, improving traffic flow and safety.

II. Related Work

In recent days, Machine Learning and Deep Learning Techniques have been evolving as a powerful toolkit in several fields, such as sustainable prawn cultivation [6], mushroom cultivation [7], efficient energy management [8] and widespread areas of study, including video dehazing.

A. Review of Existing Literature

Li et al. [9] introduced a depth-aware video dehazing technique that estimates the depth of haze present in the image. It reconstructs a clear image by comparing the depths of the haze with the other objects. Their method allows for real-time haze removal for dynamic scenes, emphasizing improved computational efficiency for real-world applications.

Zhang et al. [10] proposed the first Real-world Video Dehazing (REVIDE) dataset which can be used for supervised learning dehazing algorithms. They captured perfectly similar videos of a scene with and without haze. To effectively handle the temporal redundancy among the hazy frames, they developed a Confidence Guided and Improved Deformable Network (CG-IDN). The hazy scenes in the REVIDE dataset are more realistic than the synthetic datasets.

Xu et al. [11] presented MAP-Net, a memory-aided video dehazing network that uses physical priors and U-Net-like structures. The hazy frames are encoded and passed through two main modules, Memory-Based Physical Prior Guidance (MPG) and Multi-Range Scene Recovery (MSR). MPG updates its knowledge upon each iteration, while MSR focuses on restoring the features of the image. Outputs from each decoder layer are combined through skip connections and upsampling to produce dehazed frames. The system ensures clarity in both temporal and spatial dimensions for better video quality. The model achieves high performance by aligning multi-frame temporal features, offering significant improvements in recovering scene radiance.

Yanjie Tan et al. [12] discussed MAPD, a real-time dehazing accelerator using FPGA-based hardware. They introduced pre-defined knowledge of the atmosphere into the video dehazing accelerator (MAPD) and implemented it on a field programmable gate array (FPGA). MAPD applies heterogeneous atmospheric light in the picture to balance brightness deviation. Considering the parallel characteristics of FPGA, MAPD leverages the redundant information between

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consecutive frames to speed up the dehazing process and designs a clear image from the same.

Junkai Fan et al. [13] highlighted the misalignments brought about by spatial and temporal features in hazy datasets for autonomous driving. Their algorithm incorporates three key modules, Non-Aligned Reference Frame Matching (NRFM) for frame selection, Flow-Guided Cosine Attention Sampler (FCAS) for improved multi-frame alignment, and Deformable Cosine Attention Fusion (DCAF) for effective feature aggregation. It effectively eliminates haze while preserving brightness and texture, enhancing visibility and safety for autonomous driving.

III. Methodology

A. Dataset Overview

The dataset chosen for this study is derived from the MARS (MultiAgent, multitraveRSal, and multimodal) Dataset, released by NYU Tandon School of Engineering in collaboration with May Mobility. To specifically focus on dehazing and object detection under adverse weather conditions, we manually selected 20 video sequences from the dataset that exhibited visible haze, fog, or reduced visibility due to atmospheric interference.

Each video was processed at its native frame rate using a variable frame rate extraction module. This module first analyzed the video's properties (including frame rate, resolution, and timestamp metadata) and dynamically extracted individual frames for training and evaluation. As a result, this process yielded approximately 20,000 individual hazy image frames, forming the dataset used for training and testing our dehazing and object detection pipeline.

For the dataset split, we employed a simple random partition:

Training set: 70 % of the frames ($\tilde{1}$4,000 images)

Validation set: 15 % of the frames ($\tilde{3}$,000 images)

Testing set: 15 % of the frames ($\tilde{3}$,000 images)

The selection process ensured diversity in lighting conditions, haze densities, and scene contents (urban, suburban, and mixed traffic environments), enabling robust model evaluation under varying real-world conditions. Sample frames illustrating the hazy conditions are provided in the supplementary material to aid reproducibility.

B. Model Architecture

The system processes hazy video footage captured at 60 frames per second, treating each frame as a separate hazy image. These frames are fed into ConvIR, a neural network model specifically designed for image restoration, particularly dehazing. ConvIR operates using encoder-decoder blocks and leverages residual layers to refine multi-scale features. By merging low-level and high-level information through skip connections, ConvIR ensures the preservation of crucial image details while progressively enhancing the image clarity. This results in the generation of clearer frames that can be used for subsequent processing.

Once the dehazing is complete, the processed images are passed to YOLO (You Only Look Once), a real-time object detection model. YOLO divides each image into a grid and predicts bounding boxes along with class probabilities for objects within the scene in a single forward pass. Unlike traditional methods that perform region proposals and classification separately, YOLO treats object detection as a regression problem. This approach makes YOLO exceptionally fast and efficient, enabling it to detect multiple objects in the dehazed images swiftly. Combined, this workflow enhances real-time performance for applications requiring image clarity and object detection, such as surveillance or autonomous driving.

C. Algorithm

Algorithm 1 describes the proposed method for video dehazing and object detection using deep learning models such as ConvIR and YOLO. The input is hazy video footage processed frame by frame at the frame rate in which it was shot.The variability of the frame rate provides a non-distorted image at each frame making it simpler for dehazing. Each frame is dehazed using ConvIR, and the output is fed into the YOLO model for object detection.

D. CONVIR

ConvIR is an image restoration model that consists predominantly of two main blocks, which are supplemented by subsequent modules that help capture various details of the image at different levels. This allows efficient removal of haze at a multi-scale level. The following subsections illustrate the various components of the ConvIR model.

1) Shallow Convolution Module (SCM): Shallow Convolution Module (SCM) is the first step in the ConvIR model. It is required to perform downsampling on the input images which then helps to understand both the fine and global features at different sizes.In non-synthetic images, i.e., natural settings, the haze may impact the image differently at different parts of the image. Using SCM at different scales, the model can better analyze the small and large features independently. The SCM uses a 3X3 Convolution *Cout* filter and a padding of *P*=1. This ensures the spatial dimensions remain unchanged. Here each image is converted into feature maps where each feature map highlights certain low-level features [14], namely Edges (horizontal, vertical, diagonal), Textures (repeated patterns), Corners and blobs (key regions) and Color Gradients (intensity transitions) of the image. These feature maps are then normalized using Batch Normalization. Finally, the Relu activation function sets all the negative values in the feature map to zero which then introduces non-linearity into the module which helps understand complex patterns with ease.

The Batch Normalization transformation consists of two main steps: *Normalization* and *Affine Transformation (Scaling and Shifting)*.

- **Normalization**: Each input value x_i in the feature map is normalized to have zero mean and unit variance across a mini-batch:

$$\hat{x}_i = \frac{x_i - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}, \tag{1}$$

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Algorithm 1 Algorithm for Video Dehazing and Object Detection

```
Initialization
Initialize Models: ConvIR for dehazing, YOLO for object detection
Set Input Video Parameters: fps ← dynamic
true_positives ← 0, false_positives ← 0
true_negatives ← 0, false_negatives ← 0
all_detected_objects ← []
all_ground_truth ← []
Input Hazy Video: video ← load_video(input file)
for frame ← 1 to total_frames(video) do
    Extract Current Frame: hazy_frame ← extract_frame(video, frame)
    Apply Dehazing: dehazed_frame ← ConvIR(hazy_frame)
    Perform Object Detection: detections ← YOLO(dehazed_frame)
    Store Results:
    all_detected_objects.append(detections)
    ground_truth ← get_ground_truth(frame)
    all_ground_truth.append(ground_truth)
    Update TP, FP, TN, FN:
    for object ∈ detections do
        if object ∈ ground_truth then
            true_positives ← true_positives + 1
        else
            false_positives ← false_positives + 1
        end if
    end for
    for truth ∈ ground_truth do
        if truth ∉ detections then
            false_negatives ← false_negatives + 1
        end if
    end for
end for
Calculate Metrics:
precision ← true_positives / (true_positives + false_positives)
recall ← true_positives / (true_positives + false_negatives)
f1_score ← (2 × precision × recall) / (precision + recall)
accuracy ← (true_positives + true_negatives) / total_frames(video)
Output Metrics:
Output precision, recall, f1_score, accuracy =0
```

where:

- x_i is the input value for the i-th pixel in channel c,
- μ_c is the mean of channel c across the batch:

$$\mu_c = \frac{1}{B \cdot H \cdot W} \sum_{b=1}^{B} \sum_{h=1}^{H} \sum_{w=1}^{W} x_{bchw}, \tag{2}$$

- σ_c^2 is the variance of channel c:

$$\sigma_c^2 = \frac{1}{B \cdot H \cdot W} \sum_{b=1}^{B} \sum_{h=1}^{H} \sum_{w=1}^{W} (x_{bchw} - \mu_c)^2, \tag{3}$$

- ϵ is a small constant added for numerical stability.

- **Affine Transformation (Scaling and Shifting)**: The normalized value $\hat{x}_i$ is scaled and shifted using two learnable parameters: γ (scale) and β (shift):

$$y_i = \gamma_c \hat{x}_i + \beta_c, \tag{4}$$

where:

- γ_c is the learnable scaling parameter for channel c,
- β_c is the learnable shifting parameter for channel c,
- y_i is the output after Batch Normalization.

This process ensures that the inputs to the next layer have a consistent distribution, improving training stability and performance.

The convolution operation applies C_{out} filters of size $k \times k$ (e.g., 3×3) to the input X. The convolution output is given by:

$$F_{\text{conv}} = \text{Conv}_{k \times k}(X, W_c) + b_c, \tag{5}$$

where:

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- W_c are the learnable weights of the convolution filters,
- b_c is the learnable bias term,
- k is the kernel size (e.g., 3×3).

Batch Normalization

The convolution output F_{conv} is normalized using Batch Normalization (BN) to stabilize training:

$$F_{\text{BN}} = \frac{F_{\text{conv}} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} \cdot \gamma_c + \beta_c, \tag{6}$$

where:

- μ_c and σ_c^2 are the mean and variance of the feature map F_{conv} for channel c,
- ϵ is a small constant for numerical stability,
- γ_c and β_c are learnable scaling and shifting parameters.

ReLU Activation

The normalized feature map F_{BN} is passed through a ReLU activation function to introduce non-linearity:

$$F_{\text{shallow}} = \text{ReLU}(F_{\text{BN}}), \tag{7}$$

where:

- $\text{ReLU}(x) = \max(0, x)$.

Final Output

The final output of the Shallow Convolution Layer is:

$$F_{\text{shallow}} = \text{ReLU}\left(\frac{\text{Conv}_{k\times k}(X, W_c) + b_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} \cdot \gamma_c + \beta_c\right). \tag{8}$$

2) Encoding: Encoding is the first major block of the ConvIR model. It is responsible to take multi-level features from the feature maps processed by the SCM. It is responsible to actively refine the input feature maps.To achieve this the encoder block consists of multiple Residual Blocks(ResBlocks) which help preserve data derived from the images. Within the Encoder there are three encoding blocks. Each encoding block is responsible to extract certain specific features. Encoder Block1 is responsible for the extraction of fine grained edges and boundaries. Encoder Block2 is responsible to attain Medium Level features such as simple shapes and structures.With respect to our dataset the features would be to identify buildings, cars and it's surroundings. Encoder Block3 is responsible to analyze high level semantic features and patterns. The encoder block applies ResBlocks sequentially. Each ResBlocks consists of two *3X3* convolutions for feature extraction followed by a Relu activation function.The final ResBlock uses a Deep Pool layer that captures both local and global layers and combines them at various resolutions.This enables the field to learn more robust representation of the input by integrating multiple receptive fields.

Given an input feature map $F_{\text{in}} \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$, the Encoder Block output F_{out} is computed as follows:

$$F_{\text{out}} = \mathcal{R}_n(F_{\text{in}}), \tag{9}$$

where:

- $\mathcal{R}_n$ represents a sequence of n Residual Blocks.
- F_{in}: Input feature map.
- F_{out}: Output feature map after the Encoder Block.

Residual Block

Each Residual Block $\mathcal{R}$ processes its input $F_{\text{in}}^{(i)}$ as follows:

$$F_1^{(i)} = \text{ReLU}(\text{Conv}_{3\times 3}(F_{\text{in}}^{(i)}, W_1^{(i)})), \tag{10}$$

$$F_2^{(i)} = \text{Conv}_{3\times 3}(F_1^{(i)}, W_2^{(i)}), \tag{11}$$

$$F_{\text{out}}^{(i)} = F_{\text{in}}^{(i)} + F_2^{(i)} \quad \text{(skip connection)}, \tag{12}$$

where:

- $F_{\text{in}}^{(i)}$: Input feature map to the i-th Residual Block.
- $W_1^{(i)}, W_2^{(i)}$: Learnable weights of the two 3×3 convolution filters.
- $F_1^{(i)}$: Intermediate feature map after the first convolution and ReLU activation.
- $F_2^{(i)}$: Output feature map after the second convolution.
- $F_{\text{out}}^{(i)}$: Final output of the Residual Block after the skip connection.

Final Output

The final output F_{out} of the Encoder Block is the cumulative result of all n Residual Blocks:

$$F_{\text{out}} = \mathcal{R}_n(F_{\text{in}}), \tag{13}$$

where $\mathcal{R}_n$ is the sequence of n Residual Blocks with a filtering operation applied in the last block.

3) Feature Aggregator(FAM): Feature Aggregator Module is crucial for merging and aligning multiple features from different sources(hierarchical features from the encoder and the spatial features). This helps preserve high level features which are crucial for dehazing tasks. This helps to deal with loss of detail which is crucial for tasks dealing with removal of noise(haze in this case).This also helps in retaining finer detail during the decoding process which outputs a higher resolution image once dehazing occurs.The features are first concatenated and then passed through a *3X3* convolution matrix in order to reduce the channels and fuse the features.This aligns the channel dimensions for downstream processing.

Given two input feature maps $F_1 \in \mathbb{R}^{B\times C\times H\times W}$ and $F_2 \in \mathbb{R}^{B\times C\times H\times W}$, the output F_{out} of the Feature Aggregator Module is computed as follows:

$$F_{\text{cat}} = \text{Concat}(F_1, F_2, \text{dim} = 1), \tag{14}$$

where:

- F_1: First input feature map,
- F_2: Second input feature map,
- Concat: Concatenation operation along the channel dimension.

The concatenated feature map F_{cat} has dimensions $B \times (2C) \times H \times W$.

Convolution for Feature Merging

The concatenated feature map F_{cat} is passed through a 3×3 convolution to reduce the number of channels back to C and fuse the features:

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$$F_{\text{out}} = \text{Conv}_{3\times3}(F_{\text{cat}}, W_{\text{merge}}) + b_{\text{merge}}, \tag{15}$$

where:

- $\text{Conv}_{3\times3}$: Convolution operation with kernel size 3×3,
- W_{merge}: Learnable weights of the convolution filter,
- b_{merge}: Bias term of the convolution,
- F_{out}: Final output feature map after feature aggregation.

Final Output

The output F_{out} of the Feature Aggregator Module is a fused representation of the input features:

$$F_{\text{out}} = \text{Conv}_{3\times3}(\text{Concat}(F_1, F_2, \text{dim} = 1), W_{\text{merge}}) + b_{\text{merge}}. \tag{16}$$

4) Decoding: Decoding Block(DBlock) is used to upsample the features attained by the encoding layer. the DBlock is responsible to refine the encoded layers and reconstruct the spatial details progressively. It uses Hierarchical Upsampling to progressively restore the spatial resolution lost during encoding.The structure of the Deodding layer is similar to that of the encoding layer, the differnce being that it focuses on downsampling instead of encoding. The Resblock connections from the encoder provides additonal information to better refine the feature maps.This process is repeated across multiple Decoder Blocks, with each block progressively restoring the spatial resolution of the feature maps.High-level features are reconstructed first, followed by mid-level structures and fine-grained details, such as textures and edges, in subsequent blocks. The Decoder Block ensures that the restored image maintains both the semantic context and fine details critical for high-quality reconstruction.

Input Feature Processing

Each input feature map F_{in} is passed through a sequence of Residual Blocks (ResBlocks), which refine the features:

$$F_{\text{residual}} = F_{\text{in}} + \text{Conv}_{3\times3}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(F_{\text{in}})))). \tag{17}$$

where:

- F_{in}: Input feature map,
- F_{residual}: Refined feature map after a Residual Block,
- $\text{Conv}_{3\times3}$: Convolution with kernel size 3×3,
- BN: Batch Normalization,
- ReLU: Rectified Linear Unit activation function.

Multi-Scale Context Aggregation

In the final ResBlock of the DBlock, a Deep Pool Layer enhances features by aggregating multi-scale contextual information:

$$F_{\text{context}} = \sum_{i=1}^{N} \text{Upsample}(\text{Conv}_{3\times3}(\text{AvgPool}_{s_i}(F_{\text{residual}}), W_i)), \tag{18}$$

$$\text{size} = (H, W),$$

where:

- F_{context}: Aggregated multi-scale feature map,
- AvgPool_{s_i}: Average pooling at scale s_i,
- W_i: Learnable weights for convolution at scale s_i,
- H, W: Height and width of the original input feature map,
- N: Number of pooling scales.

Feature Fusion with Skip Connections

Features from the corresponding Encoder Block are concatenated with the upsampled output from the previous Decoder Block:

$$F_{\text{fused}} = \text{Concat}(F_{\text{encoder}}, F_{\text{upsampled}}, \text{dim} = 1), \tag{19}$$

where:

- F_{encoder}: Feature map from the Encoder,
- $F_{\text{upsampled}}$: Upsampled output from the previous Decoder Block,
- Concat: Concatenation operation along the channel dimension.

Output Refinement

The fused features are passed through a 3×3 convolution for final refinement:

$$F_{\text{out}} = \text{Conv}_{3\times3}(F_{\text{fused}}, W_{\text{final}}), \tag{20}$$

where:

- F_{out}: Refined feature map output from the Decoder Block,
- W_{final}: Learnable weights of the final convolution layer.

E. YOLO

YOLO(You Only Look Once) is a deep learning model designed for real-time object detection in images and videos. Unlike traditional object detection that predominantly relies on a sliding window in order to understand the characteristics of the object in frame YOLO reframes detection as a single regression problem, predicting bounding boxes and class probabilities directly from an image in one evaluation. The whole image is taken as an input and the entire image is then converted into an *SXS* grids. Each grid predicts a fixed number of bounding boxes, confidence scores for each box, and probabilities for object classes. YOLO's unified architecture enables it to be extremely fast while maintaining accuracy, making it suitable for applications requiring real-time performance such as autonomous driving and surveillance. The YOLO model has proven to be faster than the existing methodology, providing us high accuracy in detecting objects - cars, pedestrians,traffic lights, and stop signs from our dehazed image output. Proving to be highly beneficial while driving or for video surveillance.

F. YOLOV5

YOLOV5 is the fifth variation of the YOLO model is employed as the object detection model in our study due to its lightweight structure, real-time performance and accuracy, which are critical factors for deploying systems that require immediate feedback, such as driver-assistance technologies or autonomous vehicles.YOLOV5's backbone, CSPNet(Cross-Stage Partial Network) efficient features extraction while its PANet(Path Aggregator Network) enhances multi-scale fusion which is crucial for detecting vehicles of various sizes especially in challenging visible conditions. YOLOV5 model offers

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support for advanced data augmentation technique, including mosaic augmentation, which is beneficial in improving the robustness of detection in diverse environmental environments. The availability of smaller model variants such as YOLOV5s allows integration with vehicle dashcams, which have limited computational power. This setup would allow real time detection of nearby vehicles providing timely warnings to drivers or the autonomous driving system.

1. Bounding Box Prediction

$$
\begin{aligned}
x &= \sigma(t_x) + c_x, \\
y &= \sigma(t_y) + c_y, \\
w &= p_w e^{t_w}, \\
h &= p_h e^{t_h},
\end{aligned}
$$

Where:

- x, y: Coordinates of the bounding box center.
- w, h: Width and height of the bounding box.
- t_x, t_y, t_w, t_h: Predicted box offsets from the neural network.
- c_x, c_y: Top-left corner of the grid cell containing the object.
- p_w, p_h: Dimensions (width and height) of the anchor box.
- $\sigma(x)$: Sigmoid activation function to normalize coordinates to the range $[0, 1]$.

2. Objectness Score

$$\text{Confidence} = \sigma(o),$$

Where:

- Confidence: Probability that an object is present in the bounding box.
- o: Raw output for the objectness score from the model.
- $\sigma(o)$: Sigmoid activation function to constrain the score between $[0, 1]$.

3. Loss Function

$$\mathcal{L} = \lambda_{\text{loc}} \mathcal{L}_{\text{loc}} + \lambda_{\text{obj}} \mathcal{L}_{\text{obj}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}},$$

Where:

- $\mathcal{L}$: Total loss for training the YOLO model.
- $\mathcal{L}_{\text{loc}}$: Localization loss, measures the error in predicted bounding box coordinates (e.g., CIoU or GIoU loss).
- $\mathcal{L}_{\text{obj}}$: Objectness loss, evaluates how well the model predicts the presence of an object in a bounding box (binary cross-entropy loss).
- $\mathcal{L}_{\text{cls}}$: Classification loss, measures the error in class probability predictions (binary cross-entropy loss).
- $\lambda_{\text{loc}}, \lambda_{\text{obj}}, \lambda_{\text{cls}}$: Weighting factors that balance the contributions of each loss component to the total loss.

Fig. 1. Primary Stage of dehazing Process. Complete haze over image.

Fig. 2. The Global environment is initially dehazed and then loaded in as the next primary image to the model

IV. Results

The combination of CONVIR and YOLOV5 demonstrates exceptional synergy and efficacy in addressing the challenges faced while driving in low visibility conditions due to atmospheric haze.CONVIR's advanced and efficient dehazing capabilities converts the dashcam footage into dehazed images enabling clarity and contrast in the dashcam footage. This enhanced visibility significantly boosts the performance of the YOLOV5 model by providing clearer inputs which then allows better confidence levels and object classification especially for objects further away from the source. This is outlined in our comparison between object detection on hazed vs dehazed images. The quantitative analysis also showcases the accuracy of 0.996 with the object detection model and also have a low PSNR value of 34.5 in the dehazing aspect of the model.The demonstrated robustness of this system underscores its potential as a practical solution for improving road safety in adverse weather conditions, ensuring reliable object detection even when visibility is significantly impaired.

A. Performance Comparison

The following table compares PSNR, SSIM, parameters, and FLOPs of various dehazing methods. Our proposed ConvIR method demonstrates competitive results.

The table compares different dehazing models based on the following metrics:

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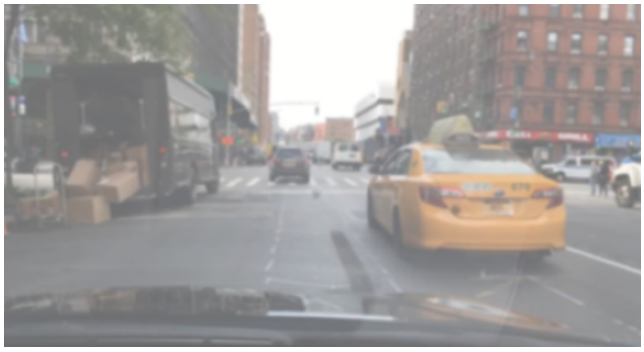

Fig. 3. Finer details are identified in relation with the environment and dehazing process is then applied

Fig. 4. A clearer image is obtained after the three stages of dehazing

Fig. 5. Object detection is performed on the clear image. This data could then help the autonomous system better detect it surroundings.

TABLE I
COMPARISON OF DIFFERENT DEHAZING METHODS [15]

Method	*PSNR*	*SSIM*	*Params (M)*	*FLOPs (G)*
DehazeNet	19.12	0.84	0.01	0.58
AOD-Net	17.15	0.83	0.002	0.12
GridDehazeNet	23.29	0.93	0.956	21.5
FFA-Net	26.96	0.95	4.456	287.8
CasDyF-Net	35.73	0.99	6.23	40.55
ConvIR (Proposed)	**34.50**	**0.99**	**14.83**	**129.34**

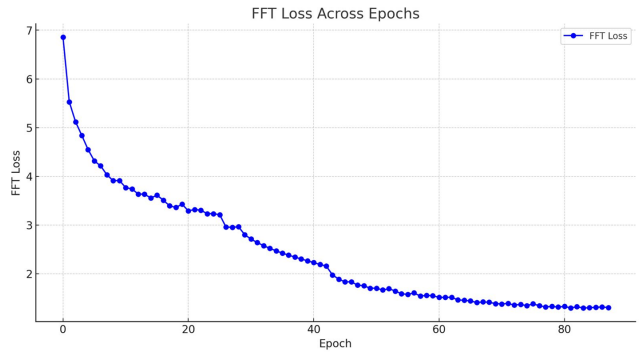

Fig. 6. Model Performance on Loss vs Epoch

- **PSNR (Peak Signal-to-Noise Ratio)**: Measures the quality of image reconstruction.
- **SSIM (Structural Similarity Index)**: Evaluates the similarity between the dehazed and hazed image. A value closer to 1 means better perceptual quality.
- **Params (M) (Number of Parameters in Millions)**: Represents the model size in terms of trainable parameters, affecting storage and computational efficiency.
- **FLOPs (G) (Floating Point Operations in Gigaflops)**: Indicates computational complexity. Higher values require more processing power.

From the table, **ConvIR (proposed)** achieves the highest PSNR (34.50) and SSIM (0.99), indicating superior dehazing quality, but at the cost of increased computational complexity.

TABLE II
COMPARISON OF DIFFERENT OBJECT DETECTION METHODS [16] [17]

Method	*Accuracy*	*Precision*	*Recall*
YOLOv5	88.3%	87.1%	87.9%
YOLOv10	89%	86%	88%
YOLOv4	86.5%	84.3%	85.1%
Faster-RNN	84.7%	82.9%	83.4%
YOLOv5 (Proposed)	**91.5%**	**89.2%**	**90.7%**

The results indicate that the proposed YOLOv5 model, when applied to dehazed video frames, achieves an accuracy of 91.5%, a precision of 89.2%, and a recall of 90.7%. This improvement highlights the enhanced performance of the object detection stage due to effective preprocessing by the dehazing module.

V. CONCLUSION

This study demonstrates how to optimize object detection systems by integrating them with robust video dehazing systems. ConvIR's encoder-decoder architecture with residual layers effectively removes haze from video frames, delivering clearer images that significantly enhance the performance of subsequent object detection. The dehazed outputs are processed using YOLOv5, which treats object detection as a regression problem. YOLOv5's efficient grid-based approach enables swift and precise identification of multiple objects within a scene, even in low-visibility scenarios.

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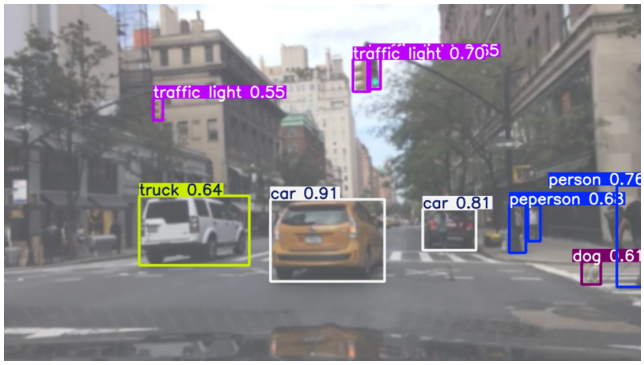

Fig. 7. YOLOv5 prediction on hazed image

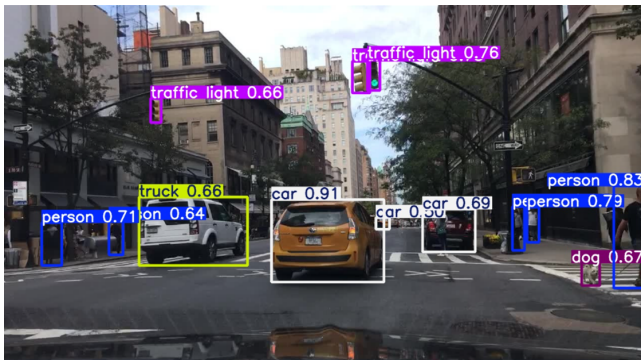

Fig. 8. YOLOv5 prediction on dehazed image

The results validate the effectiveness of our integrated system, which combines ConvIR-based dehazing with YOLOv5 object detection. The overall object detection accuracy is significantly improved to 0.996, with a PSNR value of 34.5 in the dehazing stage. This improvement in accuracy is a result of embedding the object detection model with our dehazing algorithm. A comparative analysis of hazed and dehazed frames demonstrates substantial gains in visibility and detection confidence, especially for distant objects. This improvement directly translates to enhanced safety and reliability in real-world applications such as autonomous driving, where accurate and timely object detection is critical.

The implications of this work extend beyond autonomous driving to other domains like surveillance and environmental monitoring. By improving image clarity and object detection reliability, the system contributes to overcoming visibility-related challenges in computer vision. Moreover, the architecture's scalability and adaptability make it suitable for integration into diverse applications requiring real-time performance. Future work could focus on further optimizing the system for real-time deployment, incorporating additional deep learning techniques to handle other forms of environmental noise, and extending its capabilities to multi-camera systems for enhanced spatial coverage. By fostering advancements in video dehazing and object detection, this research paves the way for safer, more reliable vision-based systems in a range of critical applications.

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